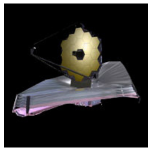

# 11094 - Linking protoplanetary disks to the Solar System: Tracing the chemical evolution of ices

Cycle: 5, Proposal Category: GO

**INVESTIGATORS**


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**OBSERVATIONS**

| *Folder* | *Observation* | *Label* | *Observing Template* | *Science Target* |
|---|---|---|---|---|
| AB Aur | | | | |
| | 1 | ABAur IFU SPEC | NIRSpec IFU Spectroscopy | (1) ABAur |
| | 2 | ABAur PSF CAL | NIRSpec IFU Spectroscopy | (3) 14Aur-PSFcal-ABAur |
| HD142527 | | | | |
| | 3 | HD142527 IFU SPEC | NIRSpec IFU Spectroscopy | (5) HD-142527 |

| *Folder* | *Observation* | *Label* | *Observing Template* | *Science Target* |
|---|---|---|---|---|
| | 4 | HD142527 PSF CAL | NIRSpec IFU Spectroscopy | (7) HD141687-PSFcal-HD142527 |


**ABSTRACT**

Volatile ices are the dominant solid components in the cold outer regions of protoplanetary disks and serve as the fundamental building blocks of planets. Yet, their compositions and spatial distributions remain poorly constrained due to the lack of spatially resolved infrared spectroscopy of the disks. We propose NIRSpec IFU observations of two well-studied protoplanetary disks, AB Aur and HD 142527, to obtain the first spatially resolved infrared reflection spectra of volatile ices in disk surface layers. These observations will enable us to detect and characterize multiple ice species, such as H2O, CO2, CO, and organic-related features and to infer their abundances using radiative-transfer modeling. Also, by comparing shadowed and illuminated regions within each disk, we will also investigate how stellar UV irradiation affects ice survival and processing. Ultimately, our program will test whether the compositional diversity observed among Kuiper Belt Objects originates from a radial gradient of ice chemistry in their natal protoplanetary disks, thus linking the chemical evolution from planet-forming disks to the Solar System.


**OBSERVING DESCRIPTION**

We propose to obtain NIRSpec IFU observations of the protoplanetary disk around AB Aur and HD142527 to trace the ice absorption distribution in the disk. Our primary target ice is H2O (3.1 µm), CO2 (4.26 µm), and CO (4.67 µm). Also, a 2.7-um OH dangling bond is key for discerning the ice mixture state. Thus we plan to use two disperser-filter combinations of G235H-F170LP and G395H-F290LP to cover 1.66-3.17µm and 2.87-5.27 µm, respectively, with R~2700. The choice of the highest spectral resolution is not to resolve the ice absorption features themselves but to mitigate the effect of detector saturation around the bright central star and the inner disk.

To see the faint structure around the very bright compact central source, we will make PSF (point spread function) reference observations to subtract the central source's halo and speckles patterns. Also, because all of the targets and PSF calibration stars are too bright for the WATA or MSATA target acquisition process, we first perform target acquisition using a nearby faint source and then apply an offset from it to place our science target and its PSF reference star in the right position (a WATA blind offset).

Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | ABAur | RA: 04 55 45.8509 (73.9410454d)<br>Dec: +30 33 3.91 (30.55109d)<br>Equinox: J2000 | Proper Motion RA: 4.01844124078382 mas/yr<br>Proper Motion Dec: -24.02653395985248 mas/yr<br>Parallax: 0.0064127"<br>Epoch of Position: 2016 | |
| *Comments: Gaia DR3 source_id: 156917493449670656<br>Category=Star<br>Description=[A stars, Protoplanetary disks]<br>Extended=YES* | | | | |
| (2) | ABAur-WATA | RA: 04 55 50.7706 (73.9615442d)<br>Dec: +30 32 51.21 (30.54756d)<br>Equinox: J2000 | Proper Motion RA: 0.03066769846598 mas/yr<br>Proper Motion Dec: -4.53567414904453 mas/yr<br>Parallax: 0.0006276"<br>Epoch of Position: 2016 | |
| *Comments: Gaia DR3 source_id: 156916771895164160<br>2MASS J : 13.929<br>Category=Unidentified<br>Description=[Infrared sources]<br>Extended=NO* | | | | |
| (3) | 14Aur-PSFcal-ABAur | RA: 05 15 24.3622 (78.8515092d)<br>Dec: +32 41 15.56 (32.68766d)<br>Equinox: J2000 | Proper Motion RA: -25.25634579770649 mas/yr<br>Proper Motion Dec: 12.33915090041149 mas/yr<br>Parallax: 0.0120126"<br>Epoch of Position: 2016 | |
| *Comments: Gaia DR3 source_id: 180702438220217344<br>Category=Star<br>Description=[A stars]<br>Extended=NO* | | | | |
| (4) | 14Aur-WATA | RA: 05 15 22.9692 (78.8457050d)<br>Dec: +32 41 52.47 (32.69791d)<br>Equinox: J2000 | Proper Motion RA: 1.01784486953215 mas/yr<br>Proper Motion Dec: -3.18953434645219 mas/yr<br>Parallax: 0.0002909"<br>Epoch of Position: 2016 | |
| *Comments: Gaia DR3 source_id: 180702502645705344<br>2MASS J : 13.953<br>Category=Unidentified<br>Description=[Infrared sources]* | | | | |
| (5) | HD-142527 | RA: 15 56 41.8725 (239.1744688d)<br>Dec: -42 19 23.67 (-42.32324d)<br>Equinox: J2000 | Proper Motion RA: -10.92376545962579 mas/yr<br>Proper Motion Dec: -26.14881451351207 mas/yr<br>Parallax: 0.0062791"<br>Epoch of Position: 2016 | |
| *Comments: Gaia DR3 source_id: 5994826707951507200<br>Category=Star<br>Description=[F stars, Protoplanetary disks]<br>Extended=YES* | | | | |
| (6) | HD142527-WATA | RA: 15 56 42.9785 (239.1790771d)<br>Dec: -42 20 13.42 (-42.33706d)<br>Equinox: J2000 | Proper Motion RA: 3.79962663386329 mas/yr<br>Proper Motion Dec: 0.00736007537435 mas/yr<br>Parallax: 0.0008931"<br>Epoch of Position: 2016 | |
| *Comments: Gaia DR3 source_id: 5994826604872280832<br>2MASS J : 12.919<br>Category=Unidentified<br>Description=[Infrared sources]<br>Extended=NO* | | | | |

| | | | | |
|---|---|---|---|---|
| (7) | HD141687-PSFcal-HD142527 | RA: 15 52 13.1426 (238.0547608d)<br>Dec: -45 17 3.62 (-45.28434d)<br>Equinox: J2000 | Proper Motion RA: -12.83480657370764 mas/yr<br>Proper Motion Dec: 4.79224966220877 mas/yr<br>Parallax: 0.0041352"<br>Epoch of Position: 2016 | |
| *Comments: Gaia DR3 source_id: 5988396386008761600<br>Category=Star<br>Description=[G stars]<br>Extended=NO* | | | | |
| (8) | HD141687-WATA | RA: 15 52 15.2546 (238.0635608d)<br>Dec: -45 17 12.69 (-45.28686d)<br>Equinox: J2000 | Proper Motion RA: 8.54349557310039 mas/yr<br>Proper Motion Dec: -9.46294714152439 mas/yr<br>Parallax: 0.0026746"<br>Epoch of Position: 2016 | |
| *Comments: Gaia DR3 source_id: 5988396489087970560<br>2MASS J : 12.829<br>Category=Unidentified<br>Description=[Infrared sources]<br>Extended=NO* | | | | |

## Observation

**Proposal 11094, Observation 1: ABAur IFU SPEC**

Mon Jun 01 09:00:12 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRSpec IFU Spectroscopy

## Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | ABAur | RA: 04 55 45.8509 (73.9410454d) | Proper Motion RA: 4.01844124078382 mas/yr | |
| | | Dec: +30 33 3.91 (30.55109d) | Proper Motion Dec: -24.02653395985248 mas/yr | |
| | | Equinox: J2000 | Parallax: 0.0064127" | |
| | | | Epoch of Position: 2016 | |

*Comments: Gaia DR3 source_id: 156917493449670656*
*Category=Star*
*Description=[A stars, Protoplanetary disks]*
*Extended=YES*

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 ABAur-WATA | WATA | SUB32 | F110W | NRSRAPID | 3 | 1 | 1 | 0.08 | 303882 |

## Template

| HFF Readout Mode |
|---|
| false |

## Dithers

| # | Dither Type | Size | Starting Point | Number of Points | Points |
|---|---|---|---|---|---|
| 1 | CYCLING | SMALL | 1 | 16 | |

## Spectral Elements

| # | Grating/Filter | Readout Pattern | Groups/Int | Integrations/Exp | Leakcal | Dither | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | G235H/F170LP | NRSRAPID | 2 | 5 | false | true | NONE | 16 | 80 | 2576.825 | 276570 |
| 2 | G395H/F290LP | NRSRAPID | 2 | 5 | false | true | NONE | 16 | 80 | 2576.825 | 276570 |

| Special Requirements | |
|---|---|
| | Aperture PA Range 213.97164917 to 238.97164917 Degrees (V3 75.0 to 100.0)<br><br>Group Observations 1, 2, Non-interruptible |

## Observation

**Proposal 11094, Observation 2: ABAur PSF CAL**

Mon Jun 01 09:00:12 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRSpec IFU Spectroscopy

## Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (3) | 14Aur-PSFcal-ABAur | RA: 05 15 24.3622 (78.8515092d)<br>Dec: +32 41 15.56 (32.68766d)<br>Equinox: J2000 | Proper Motion RA: -25.25634579770649 mas/yr<br>Proper Motion Dec: 12.33915090041149 mas/yr<br>Parallax: 0.0120126"<br>Epoch of Position: 2016 | |

*Comments: Gaia DR3 source_id: 180702438220217344*
*Category=Star*
*Description=[A stars]*
*Extended=NO*

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 14Aur-WATA | WATA | SUB32 | F110W | NRSRAPID | 3 | 1 | 1 | 0.08 | 303882 |

## Template

| HFF Readout Mode |
|---|
| false |

## Dithers

| # | Dither Type | Size | Starting Point | Number of Points | Points |
|---|---|---|---|---|---|
| 1 | CYCLING | SMALL | 1 | 16 | |

## Spectral Elements

| # | Grating/Filter | Readout Pattern | Groups/Int | Integrations/Exp | Leakcal | Dither | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | G235H/F170LP | NRSRAPID | 2 | 5 | false | true | NONE | 16 | 80 | 2576.825 | 276570 |
| 2 | G395H/F290LP | NRSRAPID | 2 | 5 | false | true | NONE | 16 | 80 | 2576.825 | 276570 |

| Special Requirements | |
|---|---|
| | Group Observations 1, 2, Non-interruptible |

## Observation

**Proposal 11094, Observation 3: HD142527 IFU SPEC**

Mon Jun 01 09:00:12 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRSpec IFU Spectroscopy

## Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (5) | HD-142527 | RA: 15 56 41.8725 (239.1744688d)<br>Dec: -42 19 23.67 (-42.32324d)<br>Equinox: J2000 | Proper Motion RA: -10.92376545962579 mas/yr<br>Proper Motion Dec: -26.14881451351207 mas/yr<br>Parallax: 0.0062791"<br>Epoch of Position: 2016 | |

*Comments: Gaia DR3 source_id: 5994826707951507200*
*Category=Star*
*Description=[F stars, Protoplanetary disks]*
*Extended=YES*

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 6 HD142527-WATA | WATA | SUB32 | F110W | NRSRAPID | 3 | 1 | 1 | 0.08 | 303882 |

## Template

| HFF Readout Mode |
|---|
| false |

## Dithers

| # | Dither Type | Size | Starting Point | Number of Points | Points |
|---|---|---|---|---|---|
| 1 | CYCLING | SMALL | 1 | 16 | |

## Spectral Elements

| # | Grating/Filter | Readout Pattern | Groups/Int | Integrations/Exp | Leakcal | Dither | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | G235M/F170LP | NRSRAPID | 2 | 5 | false | true | NONE | 16 | 80 | 2576.825 | 276570 |
| 2 | G395M/F290LP | NRSRAPID | 2 | 5 | false | true | NONE | 16 | 80 | 2576.825 | 276570 |

| | |
|---|---|
| **Special Requirements** | Aperture PA Range 233.97164917 to 238.97164917 Degrees (V3 95.0 to 100.0)<br><br>Group Observations 3, 4, Non-interruptible |

## Observation

**Proposal 11094, Observation 4: HD142527 PSF CAL**

Mon Jun 01 09:00:12 GMT 2026

**Diagnostic Status: Warning**

Observing Template: NIRSpec IFU Spectroscopy

## Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (7) | HD141687-PSFcal-HD142527 | RA: 15 52 13.1426 (238.0547608d)<br>Dec: -45 17 3.62 (-45.28434d)<br>Equinox: J2000 | Proper Motion RA: -12.83480657370764 mas/yr<br>Proper Motion Dec: 4.79224966220877 mas/yr<br>Parallax: 0.0041352"<br>Epoch of Position: 2016 | |

*Comments: Gaia DR3 source_id: 5988396386008761600*
*Category=Star*
*Description=[G stars]*
*Extended=NO*

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 8 HD141687-WATA | WATA | SUB32 | F110W | NRSRAPID | 3 | 1 | 1 | 0.08 | 303882 |

## Template

| HFF Readout Mode |
|---|
| false |

## Dithers

| # | Dither Type | Size | Starting Point | Number of Points | Points |
|---|---|---|---|---|---|
| 1 | CYCLING | SMALL | 1 | 16 | |

## Spectral Elements

| # | Grating/Filter | Readout Pattern | Groups/Int | Integrations/Exp | Leakcal | Dither | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | G235M/F170LP | NRSRAPID | 2 | 5 | false | true | NONE | 16 | 80 | 2576.825 | 276570 |
| 2 | G395M/F290LP | NRSRAPID | 2 | 5 | false | true | NONE | 16 | 80 | 2576.825 | 276570 |

| Special Requirements | Group Observations 3, 4, Non-interruptible |
| --- | --- |